AI Engineering

Three tokens:

$t1: [1\ 0\ 1]$, $t2: [0\ 1\ 0]$, $t3: [1\ 1\ 0]$

$$W_Q = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} \quad W_K = \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \quad W_V = \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 0 & 1 \end{bmatrix}$$

$$X = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$$

$$Q = X \cdot W_Q \rightarrow \begin{bmatrix} 2 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$$

$$K = X \cdot W_K \rightarrow \begin{bmatrix} 0 & 2 \\ 1 & 0 \\ 1 & 1 \end{bmatrix}$$

$$V = X \cdot W_V \rightarrow \begin{bmatrix} 1 & 1 \\ 0 & 2 \\ 1 & 2 \end{bmatrix}$$

CamScanner

$$QK^T = \begin{bmatrix} 2 & 0 \\ 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 1 \\ 2 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 2 & 2 \\ 2 & 0 & 1 \\ 2 & 1 & 2 \end{bmatrix}$$

$$\frac{QK^T}{\sqrt{d_k}} \quad \hookrightarrow \sqrt{2} = \begin{bmatrix} 0 & 1.4142 & 1.4142 \\ 1.4142 & 0 & 0.7071 \\ 1.4142 & 0.7071 & 1.4142 \end{bmatrix}$$

applying row-wise softmax on $\frac{QK^T}{\sqrt{d_k}}$

$$\text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) = \underbrace{\begin{bmatrix} 0.1084 & 0.4458 & 0.4458 \\ 0.5760 & 0.1400 & 0.2840 \\ 0.4011 & 0.1978 & 0.4011 \end{bmatrix}}_{A}$$

$$\text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) \cdot V = A \cdot \begin{bmatrix} 1 & 1 \\ 0 & 2 \\ 1 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 0.5542 & 1.8916 \\ 0.8600 & 1.4240 \\ 0.8022 & 1.5989 \end{bmatrix}$$

without mask

With Masking:

Mask Matrix = $\begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

The places with ones are masked.

We'll mask the attention scores given by QK^T

So, $\frac{1}{\sqrt{2}} \begin{bmatrix} 0 & -\infty & -\infty \\ 2 & 0 & -\infty \\ 2 & 1 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & -\infty & -\infty \\ 1.4142 & 0 & -\infty \\ 1.4142 & 0.7071 & 1.4142 \end{bmatrix}$

blocking

Note: We use $-\infty$ since $e^{-\infty} = 0$ ~~making~~ these positions

now softmax on all rows,

$$\text{softmax}\left(\frac{\text{masked}(QK^T)}{\sqrt{d_k}}\right) = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0.8044 & 0.1956 & 0 \\ 0.4011 & 0.1978 & 0.4011 \end{bmatrix}}_{A_{masked}}$$

$$A_{masked} \cdot V = A_{masked} \begin{bmatrix} 1 & 1 \\ 0 & 2 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0.8044 & 1.1956 \\ 0.8022 & 1.5989 \end{bmatrix}$$

Causal masking prevents the model from "cheating" by looking at future tokens. e.g ~~the~~ most of the first token probabilities in this example were centered around ~~further~~ later tokens. Using causal masking, the tokens only see themselves and previous tokens ↑ to learn ~~from~~ enabling model better to predict the ↑ sequencing of the tokens in the language.